HealthConnect Final Data Science Executive Model Summary

Week 8 | Takudzwa Jeffrey Gumbo | AnalystLab Africa Batch E

Decision at a glance

FINAL DECISION: DO NOT DEPLOY THE CURRENT ADMISSION MODEL.

The leakage-safer Gradient Boosting candidate is retained as the final documented model for reproducibility and portfolio evidence, but its discrimination remains close to chance.

Measure	Final result	Interpretation
Holdout ROC-AUC	0.5189	Only slightly above chance
5-fold CV ROC-AUC	0.5237 +/- 0.0246	Weak generalisation
Holdout PR-AUC	0.1199	Close to the 11.7% positive rate
Default-threshold recall	0.0000	Missed all positive admissions in the holdout set

What changed after Week 7 validation

- Prediction point fixed at patient arrival / before initial clinical assessment.
- Diagnosis and Severity removed because their timing could leak clinical information.
- Age recalculated at appointment date.
- PriorVisits, appointment month and weekday added as time-safe features.
- FacilityID and DepartmentID retained with explicit caveats rather than interpreted as quality rankings.

Business meaning

The current data does not contain enough reliable pre-assessment signal to support operational admission prediction. The model should not be used to prioritize patients, allocate clinical resources, or make care decisions. Its appropriate use is as a documented analytical experiment demonstrating responsible model validation.

Next steps

- Obtain stronger pre-admission predictors and a fresh untouched test set.
- If appointment attendance is the intended use case, collect an explicit attendance/no-show outcome.
- Re-run the modelling lifecycle only after the target and prediction-time data are operationally defined.